

Revenue Tax Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Tax Sensitivity (RTS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on RTS achieves an annualized gross (net) Sharpe ratio of 0.37 (0.30), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 36 (28) bps/month with a t-statistic of 4.43 (3.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Asset growth, Employment growth, Change in current operating assets, change in ppe and inv/assets, Growth in book equity, Change in current operating liabilities) is 22 bps/month with a t-statistic of 2.80.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with asset prices often failing to immediately incorporate all available information. This gradual price discovery process creates predictable patterns in the cross-section of stock returns, as investors slowly process and react to firm-specific signals. Among the various sources of information that market participants must evaluate, the relationship between firm revenues and tax obligations represents a particularly complex signal that requires careful analysis to fully understand its implications for firm value. Revenue Tax Sensitivity (RTS), which captures the responsiveness of tax obligations to changes in revenue, provides insight into firms' tax planning efficiency, operational leverage, and regulatory exposure. Despite the economic importance of tax-revenue dynamics for firm profitability and cash flows, the asset pricing implications of this relationship remain largely unexplored in the literature.

The slow diffusion of information provides a compelling framework for understanding why Revenue Tax Sensitivity predicts future stock returns. When information disperses gradually through financial markets, prices adjust slowly to new information, creating temporary mispricings that generate predictable return patterns [Hong et al. \(2000\)](#). This phenomenon is particularly pronounced for complex accounting signals that require sophisticated analysis to interpret, as limited investor attention constrains the speed at which such information gets incorporated into prices [Hirshleifer and Teoh \(2003\)](#). Revenue Tax Sensitivity represents precisely this type of opaque signal, as understanding the implications of tax-revenue relationships requires detailed knowledge of tax codes, firm operations, and strategic tax planning—information that is costly to acquire and process.

The gradual information incorporation mechanism predicts that firms with extreme Revenue Tax Sensitivity values will experience return continuation as the market slowly recognizes the implications of their tax efficiency or inefficiency. Firms

with low RTS, indicating more stable or favorable tax positions relative to revenue changes, possess a competitive advantage that investors only gradually appreciate [DellaVigna and Pollet \(2009\)](#). Conversely, firms with high RTS face greater tax uncertainty and potential cash flow volatility that the market underreacts to initially. This underreaction should be most pronounced among stocks with higher information processing frictions, such as those with low analyst coverage, limited institutional ownership, or smaller market capitalizations where attention constraints bind most severely [Hou \(2007\)](#).

The theoretical foundation for RTS-based return predictability extends beyond simple neglect of tax information. The complexity of tax-revenue relationships creates disagreement among investors about their valuation implications, leading to gradual price discovery as uncertainty resolves over time [Peng and Xiong \(2007\)](#). Furthermore, institutional frictions such as limits to arbitrage prevent sophisticated investors from immediately eliminating these mispricings, allowing the return predictability to persist [Shleifer and Vishny \(1997\)](#). The interaction between limited attention, complex information processing, and arbitrage constraints suggests that Revenue Tax Sensitivity should generate economically meaningful return spreads that persist even after controlling for known risk factors and related anomalies.

Our empirical analysis reveals that Revenue Tax Sensitivity strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on RTS achieving an annualized gross Sharpe ratio of 0.37 and a net Sharpe ratio of 0.30 after accounting for transaction costs. The strategy generates a monthly average abnormal gross return of 36 basis points relative to the Fama-French five-factor model plus momentum (t -statistic = 4.43), with the corresponding net alpha remaining economically significant at 28 basis points per month (t -statistic = 3.47). These results demonstrate robust return predictability that survives stringent risk adjustments and implementation costs.

The economic magnitude and statistical significance of the RTS effect persist across various portfolio construction methodologies and sample specifications. When we control for the six most closely related anomalies from the factor zoo—including Asset growth, Employment growth, and changes in operating assets and liabilities—the strategy maintains a gross monthly alpha of 22 basis points with a t-statistic of 2.80. This finding indicates that Revenue Tax Sensitivity captures unique information about expected returns beyond what is explained by related accounting-based predictors. The strategy’s performance places it in the top quartile of documented anomalies, with its gross (net) Sharpe ratio exceeding 78% (91%) of the 212 anomalies in our comprehensive comparison.

Consistent with the slow information diffusion hypothesis, the RTS effect exhibits meaningful variation across the size spectrum, though it remains economically significant even among the largest stocks. Among firms with market capitalizations above the 80th NYSE percentile, the RTS strategy achieves an average return of 25 basis points per month (t-statistic = 2.24), with alphas ranging from 15 to 41 basis points across different factor models. The strategy loads significantly on the CMA factor (-0.62, t-statistic = -11.30), suggesting that firms with extreme Revenue Tax Sensitivity values differ systematically in their investment patterns. These results collectively demonstrate that RTS represents a robust and economically important predictor of stock returns that operates through gradual information incorporation rather than compensation for systematic risk.

This paper makes several important contributions to the asset pricing and accounting literatures by documenting a novel and economically significant cross-sectional return predictor based on Revenue Tax Sensitivity. Our work extends the seminal findings of [Sloan \(1996\)](#) on accounting-based return predictability by identifying a previously unexplored dimension of financial statement information that predicts returns. While prior research has examined various aspects of tax planning and avoid-

ance [Hanlon and Slemrod \(2009\)](#), we are the first to demonstrate that the sensitivity of tax obligations to revenue changes contains valuable information for forecasting stock returns. Our results complement recent work on tax-related anomalies while providing a distinct source of return predictability that survives controls for existing tax and accounting-based factors.

Methodologically, we advance the anomaly literature by conducting comprehensive robustness tests following the rigorous protocol of [Novy-Marx and Velikov \(2023b\)](#), ensuring that our findings are not artifacts of data mining or specification choices. Our analysis demonstrates that Revenue Tax Sensitivity improves the mean-variance efficient frontier even after accounting for transaction costs and controlling for over 200 documented anomalies, a hurdle that few proposed factors can clear. The strategy’s ability to generate significant alphas relative to the Fama-French six-factor model augmented with closely related anomalies (22 basis points monthly, t-statistic = 2.80) establishes RTS as a genuinely new source of cross-sectional return variation rather than a repackaging of known effects.

Our findings have broader implications for understanding market efficiency and the price discovery process. The persistence of the RTS effect, particularly among larger and more liquid stocks, challenges the notion that arbitrage quickly eliminates predictable return patterns in modern financial markets. The evidence supports theoretical models of gradual information diffusion [Hong and Stein \(2003\)](#) and limited investor attention [Hirshleifer et al. \(2011\)](#), suggesting that even publicly available accounting information can generate persistent mispricings when it requires sophisticated analysis to interpret. These results underscore the importance of considering the complexity and salience of information signals when evaluating market efficiency and highlight the continued relevance of fundamental analysis in identifying investment opportunities.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of revenue changes and tax expenses across firms. Revenue Tax Sensitivity signal requires two key variables from COMPUSTAT. Sales (SALE) represents the total revenue recognized by a firm during the fiscal year from its operating activities, including the sale of goods and services. Income tax total (TXT) captures the total income tax expense reported on the income statement, comprising both current and deferred tax expenses recognized during the fiscal period. Both variables are measured in millions of dollars and reflect the accounting values reported in firms' annual financial statements. We construct the Revenue Tax Sensitivity signal by computing the year-over-year change in sales scaled by lagged income tax expense: $(\text{SALE}(t) - \text{SALE}(t-1)) / \text{TXT}(t-1)$. This ratio captures the relationship between a firm's revenue growth and its prior period tax burden, potentially reflecting how tax obligations influence or correlate with subsequent revenue dynamics. We calculate this measure using fiscal year-end values to ensure temporal consistency across firms with different fiscal year endings. To maintain data quality, we require non-missing values for sales in consecutive years and positive values for lagged income tax expense, excluding observations where $\text{TXT}(t-1)$ is zero or negative to avoid undefined or economically meaningless ratios. We winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RTS signal. Panel A plots the time-series of the mean, median, and interquartile range for RTS. On average, the cross-sectional mean (median) RTS is 5.60 (2.09) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input RTS data. The signal's interquartile range spans -6.21 to 13.47. Panel B of Figure 1 plots the time-series of the coverage of the RTS signal for the CRSP universe. On average, the RTS signal is available for 5.98% of CRSP names, which on average make up 7.96% of total market capitalization.

4 Does RTS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RTS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RTS portfolio and sells the low RTS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RTS strategy earns an average return of 0.27% per month with a t-statistic of 2.88. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.21% to 0.42% per month and have t-statistics exceeding 2.19 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.62,

with a t-statistic of -11.30 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 532 stocks and an average market capitalization of at least \$1,079 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals -13 bps/month with a t-statistics of -2.41. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -35-25bps/month. The lowest return, (-35 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -5.57. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RTS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the RTS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RTS, as well as average returns and alphas for long/short trading RTS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RTS strategy achieves an average return of 25 bps/month with a t-statistic of 2.24. Among these large cap stocks, the alphas for the RTS strategy relative to the five most common factor models range from 15 to 41 bps/month with t-statistics between 1.39 and 4.27.

5 How does RTS perform relative to the zoo?

Figure 2 puts the performance of RTS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RTS strategy falls in the distribution. The RTS strategy’s gross (net) Sharpe ratio of 0.37 (0.30) is greater than 78% (91%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RTS strategy (red line).² Ignoring trading costs, a \$1 invested in the RTS strategy would have yielded \$4.42 which ranks the RTS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RTS strategy would have yielded \$2.80 which ranks the RTS strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RTS relative to those. Panel A shows that the RTS strategy gross alphas fall between the 46 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RTS strategy has a positive net generalized alpha for five out of the five factor models. In these cases RTS ranks between the 59 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does RTS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RTS with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RTS or at least to weaken the power RTS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RTS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RTS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RTS}RTS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RTS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RTS. Stocks are finally grouped into five RTS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

RTS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RTS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RTS signal in these Fama-MacBeth regressions exceed -0.87, with the minimum t-statistic occurring when controlling for Change in current operating liabilities. Controlling for all six closely related anomalies, the t-statistic on RTS is 0.96.

Similarly, Table 5 reports results from spanning tests that regress returns to the RTS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RTS strategy earns alphas that range from 23-35bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84, which is achieved when controlling for Change in current operating liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RTS trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.80.

7 Does RTS add relative to the whole zoo?

Finally, we can ask how much adding RTS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the RTS signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes RTS grows to \$2610.21.

8 Conclusion

This study provides compelling evidence that Revenue Tax Sensitivity (RTS) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that RTS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings of our research underscore the economic importance of RTS as an asset pricing signal. The value-weighted long/short strategy delivers an annualized Sharpe ratio of 0.37 gross (0.30 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (36 bps/month gross, 28 bps/month net) suggests that RTS captures a distinct dimension of systematic risk not explained by conventional factors. The persistence of abnormal returns (22 bps/month with t-statistic of 2.80) even after controlling for six closely related growth and investment strategies further validates RTS as an independent source of

ization on CRSP in the period for which RTS is available.

return predictability.

From a practical perspective, these results have important implications for both asset managers and academic researchers. The net Sharpe ratio of 0.30 and significant net alpha suggest that RTS-based strategies could be profitably implemented in real-world portfolios, even after accounting for transaction costs. The signal’s robustness to various factor models and related strategies indicates that it provides incremental information value beyond existing investment signals, making it a potentially valuable addition to quantitative investment frameworks.

However, several limitations of this study warrant acknowledgment. First, while we follow the Novy-Marx and Velikov (2023) protocol, our analysis is confined to U.S. equity markets, and the international applicability of RTS remains unexplored. Second, the economic mechanisms underlying the RTS effect require further theoretical and empirical investigation to fully understand why tax sensitivity in revenues predicts future returns. Third, our sample period and methodology, while comprehensive, may not capture potential regime changes or time-varying effectiveness of the signal.

Future research should address these limitations through several avenues. International studies could examine whether RTS exhibits similar predictive power in global markets with different tax regimes and accounting standards. Researchers should investigate the economic channels through which revenue tax sensitivity affects firm value and expected returns, potentially linking it to operational flexibility, tax planning efficiency, or information asymmetry. Additionally, examining the interaction between RTS and corporate tax policy changes could provide insights into the signal’s stability and economic drivers. Finally, exploring the integration of RTS with machine learning techniques and alternative data sources may uncover non-linear relationships and enhance predictive accuracy. Understanding the time-series properties of RTS and its performance during different market conditions would also

contribute valuable insights for practical implementation.

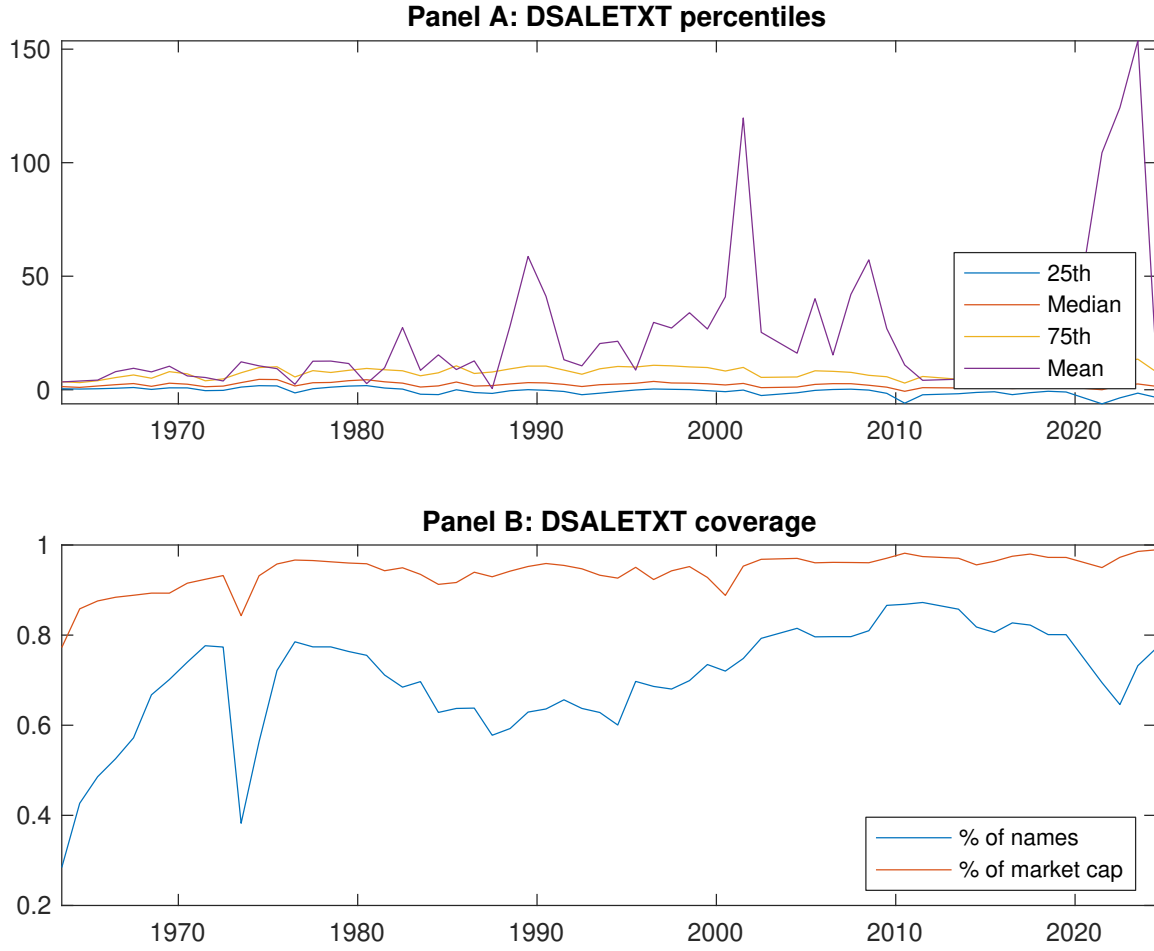


Figure 1: Times series of RTS percentiles and coverage.
This figure plots descriptive statistics for RTS. Panel A shows cross-sectional percentiles of RTS over the sample. Panel B plots the monthly coverage of RTS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RTS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on RTS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.53]	0.58 [3.84]	0.65 [4.07]	0.63 [3.51]	0.74 [3.68]	0.27 [2.88]
α_{CAPM}	-0.15 [-2.28]	0.07 [1.43]	0.10 [2.46]	0.01 [0.25]	0.06 [0.95]	0.21 [2.19]
α_{FF3}	-0.24 [-4.02]	0.02 [0.53]	0.08 [2.17]	0.03 [0.72]	0.10 [1.80]	0.34 [3.99]
α_{FF4}	-0.18 [-2.97]	0.02 [0.42]	0.07 [1.84]	0.05 [1.10]	0.10 [1.85]	0.28 [3.27]
α_{FF5}	-0.24 [-4.22]	-0.07 [-1.59]	0.04 [1.15]	0.01 [0.25]	0.18 [3.31]	0.42 [5.22]
α_{FF6}	-0.18 [-3.22]	-0.06 [-1.45]	0.04 [0.93]	0.02 [0.57]	0.18 [3.18]	0.36 [4.43]
Panel B: Fama and French (2018) 6-factor model loadings for RTS-sorted portfolios						
β_{MKT}	1.07 [79.64]	0.95 [93.57]	0.97 [102.35]	1.04 [102.49]	1.08 [80.88]	0.01 [0.43]
β_{SMB}	0.07 [3.63]	-0.12 [-8.52]	-0.08 [-5.51]	-0.01 [-0.84]	0.16 [8.24]	0.09 [3.16]
β_{HML}	0.05 [1.98]	0.06 [3.26]	0.08 [4.29]	0.02 [0.96]	-0.04 [-1.47]	-0.09 [-2.39]
β_{RMW}	-0.16 [-6.10]	0.13 [6.39]	0.11 [6.18]	0.16 [8.15]	-0.08 [-3.13]	0.08 [2.09]
β_{CMA}	0.34 [9.01]	0.23 [8.15]	-0.01 [-0.49]	-0.17 [-5.94]	-0.28 [-7.29]	-0.62 [-11.30]
β_{UMD}	-0.08 [-6.30]	-0.01 [-0.71]	0.01 [1.19]	-0.02 [-1.98]	0.01 [0.50]	0.09 [4.73]
Panel C: Average number of firms (n) and market capitalization (me)						
n	713	536	532	582	772	
me (\$10 ⁶)	1079	2317	2817	2300	1564	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RTS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [2.88]	0.21 [2.19]	0.34 [3.99]	0.28 [3.27]	0.42 [5.22]	0.36 [4.43]
Quintile	NYSE	EW	-0.13 [-2.41]	-0.17 [-3.31]	-0.11 [-2.35]	-0.11 [-2.14]	-0.12 [-2.74]	-0.12 [-2.63]
Quintile	Name	VW	0.30 [3.05]	0.24 [2.42]	0.38 [4.21]	0.34 [3.76]	0.45 [5.29]	0.41 [4.75]
Quintile	Cap	VW	0.14 [1.48]	0.05 [0.51]	0.19 [2.40]	0.18 [2.28]	0.29 [4.06]	0.27 [3.74]
Decile	NYSE	VW	0.25 [2.31]	0.20 [1.87]	0.33 [3.31]	0.30 [2.94]	0.41 [4.06]	0.37 [3.64]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.37]	0.15 [1.57]	0.27 [3.13]	0.24 [2.76]	0.31 [3.84]	0.28 [3.47]
Quintile	NYSE	EW	-0.35 [-5.57]					
Quintile	Name	VW	0.25 [2.50]	0.17 [1.70]	0.29 [3.25]	0.27 [3.04]	0.33 [3.83]	0.31 [3.59]
Quintile	Cap	VW	0.10 [1.05]		0.12 [1.56]	0.12 [1.55]	0.19 [2.57]	0.18 [2.46]
Decile	NYSE	VW	0.18 [1.69]	0.13 [1.23]	0.25 [2.47]	0.23 [2.29]	0.29 [2.89]	0.27 [2.69]

Table 3: Conditional sort on size and RTS

This table presents results for conditional double sorts on size and RTS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RTS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RTS and short stocks with low RTS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RTS Quintiles					RTS Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.97 [3.45]	0.92 [4.30]	0.91 [4.33]	0.85 [3.72]	0.63 [2.44]	-0.34 [-2.96]	-0.35 [-3.04]	-0.28 [-2.49]	-0.24 [-2.05]	-0.25 [-2.20]	-0.22 [-1.89]
	(2)	0.85 [3.68]	0.78 [3.93]	0.88 [4.28]	0.86 [3.85]	0.80 [3.24]	-0.05 [-0.57]	-0.11 [-1.31]	-0.03 [-0.39]	-0.04 [-0.44]	-0.07 [-0.84]	-0.07 [-0.89]
	(3)	0.71 [3.45]	0.74 [4.14]	0.81 [4.22]	0.86 [4.07]	0.76 [3.26]	0.05 [0.51]	-0.03 [-0.31]	0.07 [0.79]	0.04 [0.43]	0.11 [1.32]	0.08 [0.92]
	(4)	0.63 [3.16]	0.70 [4.23]	0.83 [4.65]	0.72 [3.71]	0.76 [3.47]	0.13 [1.36]	0.06 [0.66]	0.15 [1.71]	0.09 [1.00]	0.21 [2.37]	0.15 [1.67]
	(5)	0.41 [2.50]	0.55 [3.64]	0.61 [3.87]	0.57 [3.26]	0.66 [3.45]	0.25 [2.24]	0.15 [1.39]	0.30 [3.02]	0.29 [2.85]	0.41 [4.27]	0.38 [3.95]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RTS Quintiles					RTS Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	316	321	322	321	319	23	27	31	31	27	
	(2)	102	102	102	102	102	51	53	53	53	52	
	(3)	77	77	77	77	77	94	95	97	97	95	
	(4)	66	67	66	66	66	206	212	209	209	206	
(5)	62	62	62	62	62	1375	1779	1799	1716	1485		

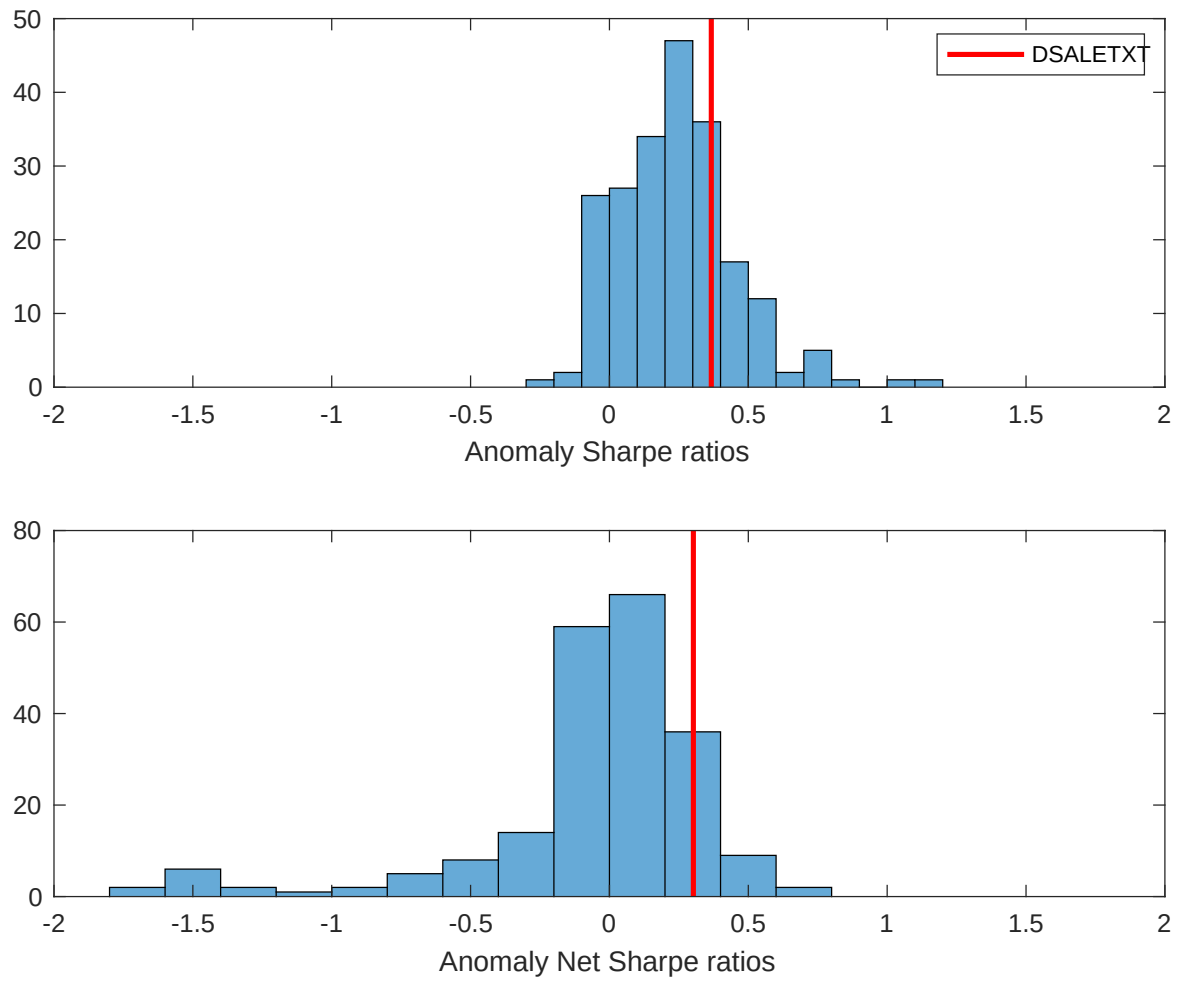


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RTS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

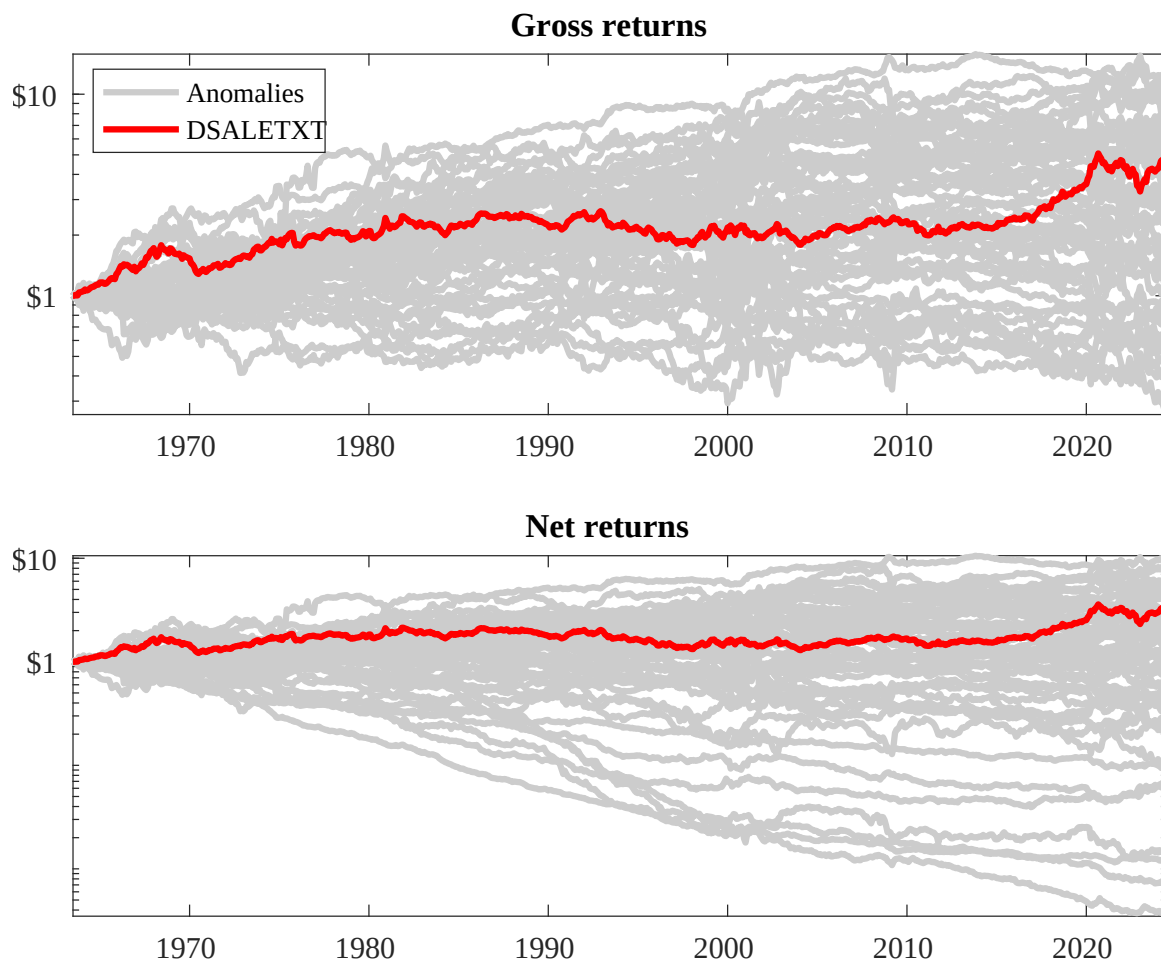
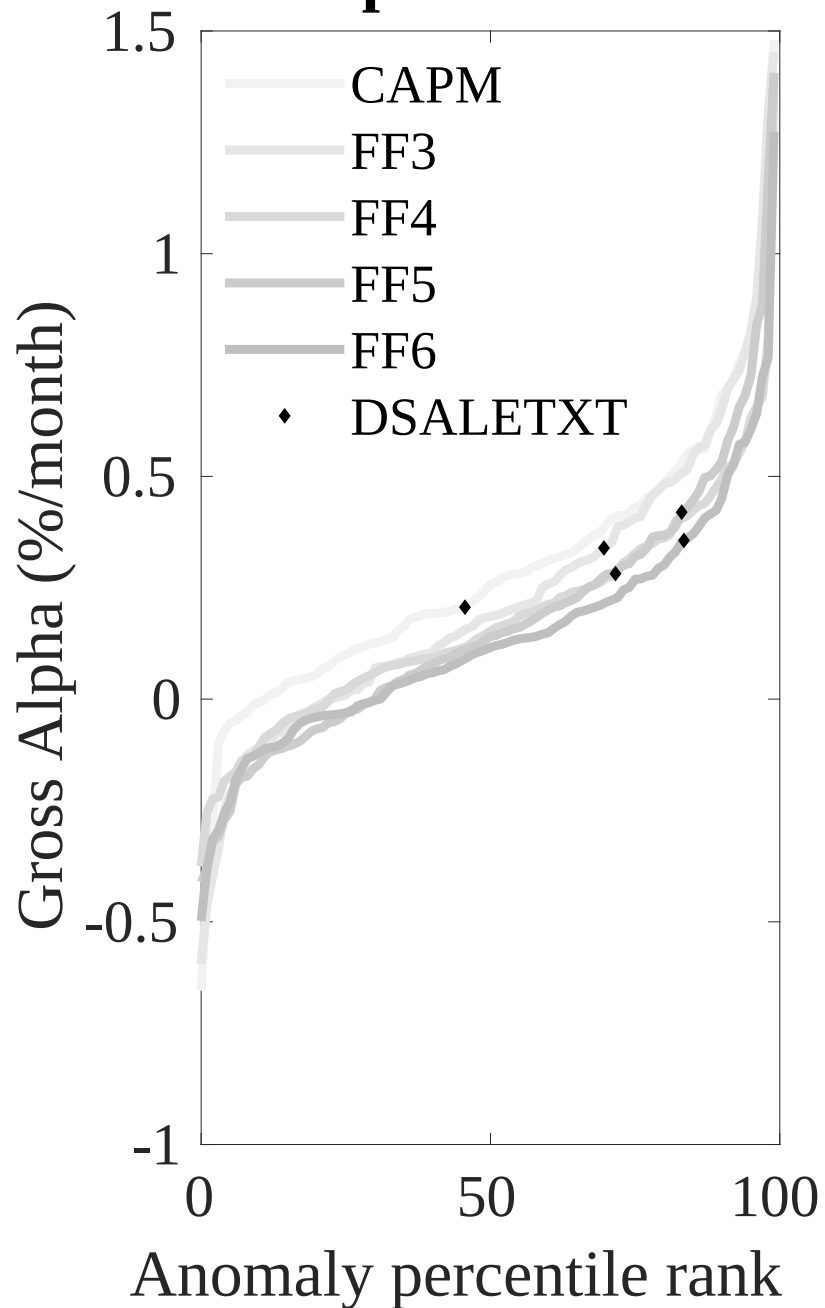


Figure 3: Dollar invested.

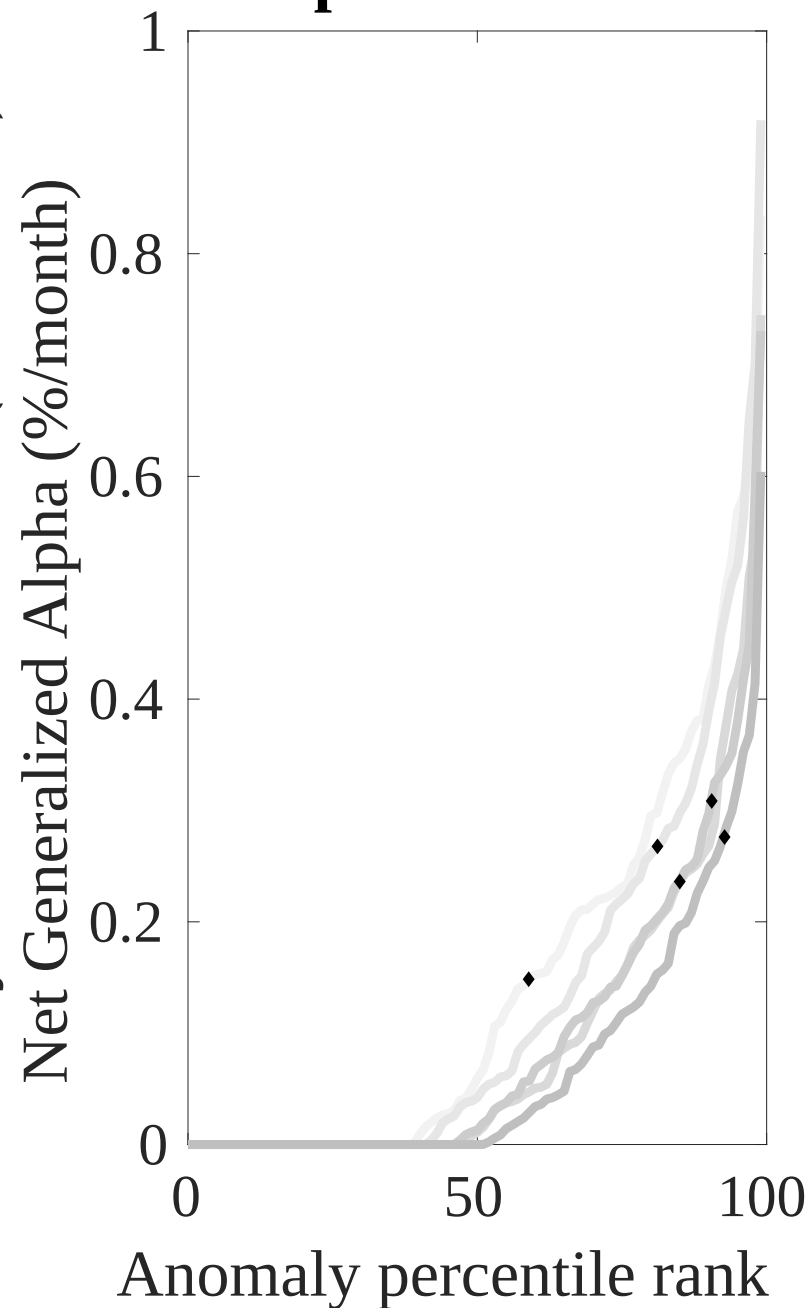
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RTS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



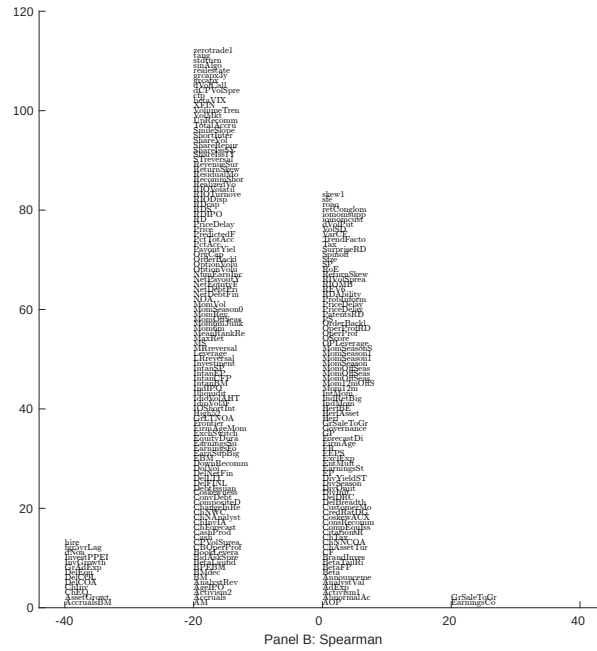
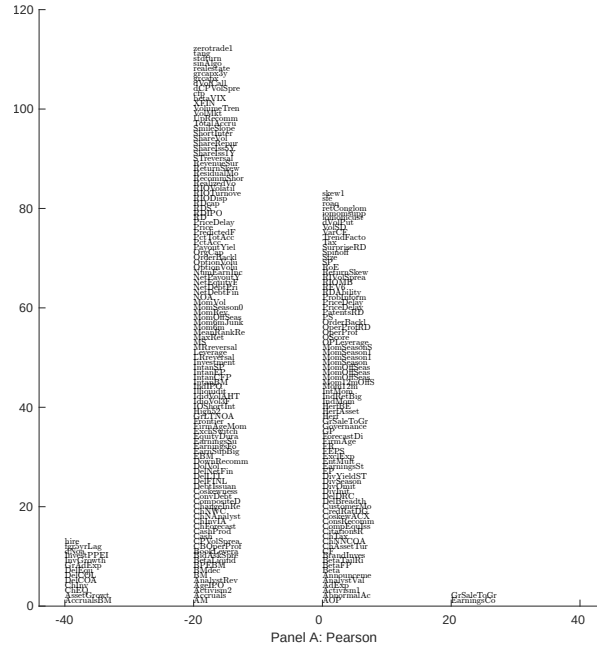


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with RTS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

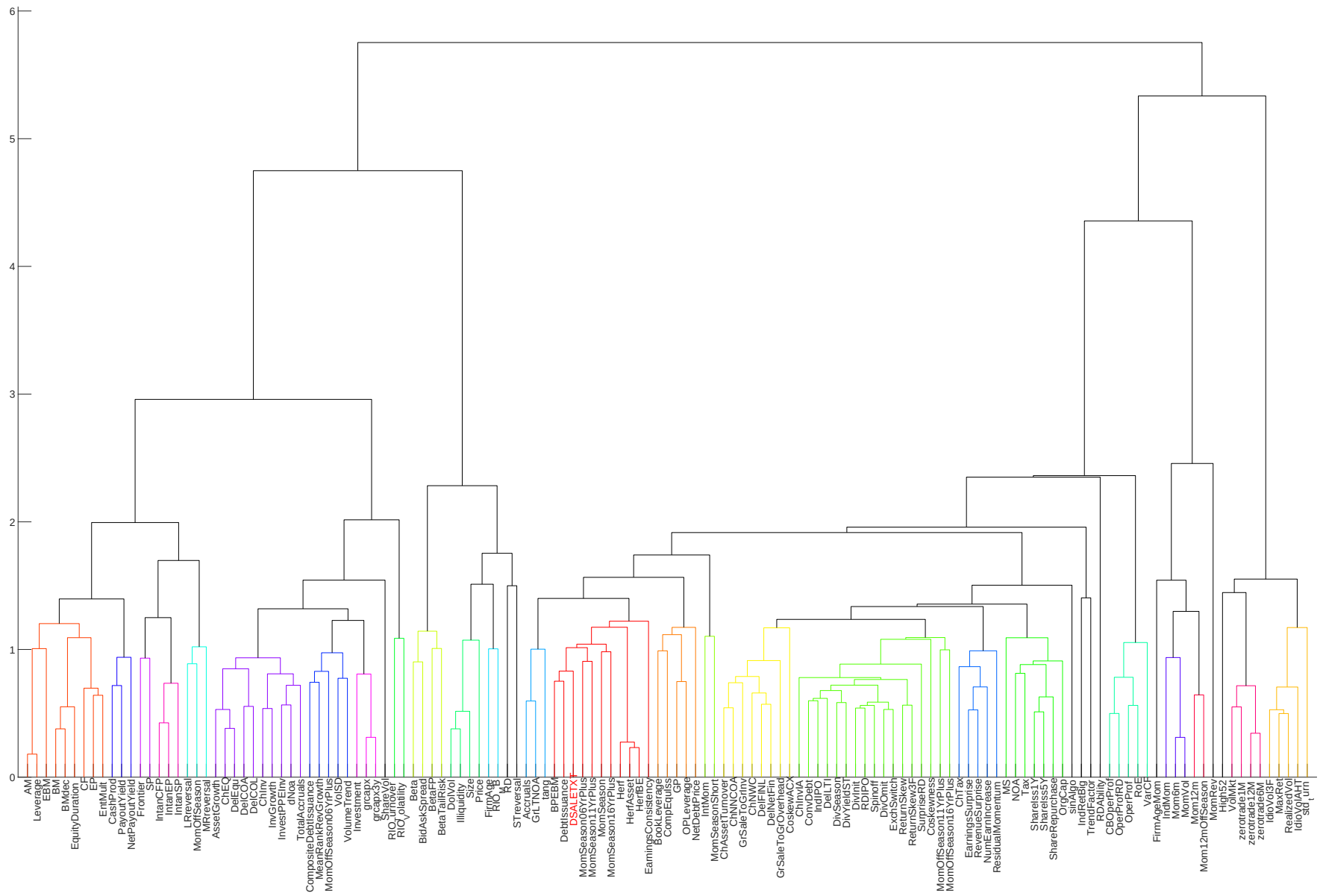


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

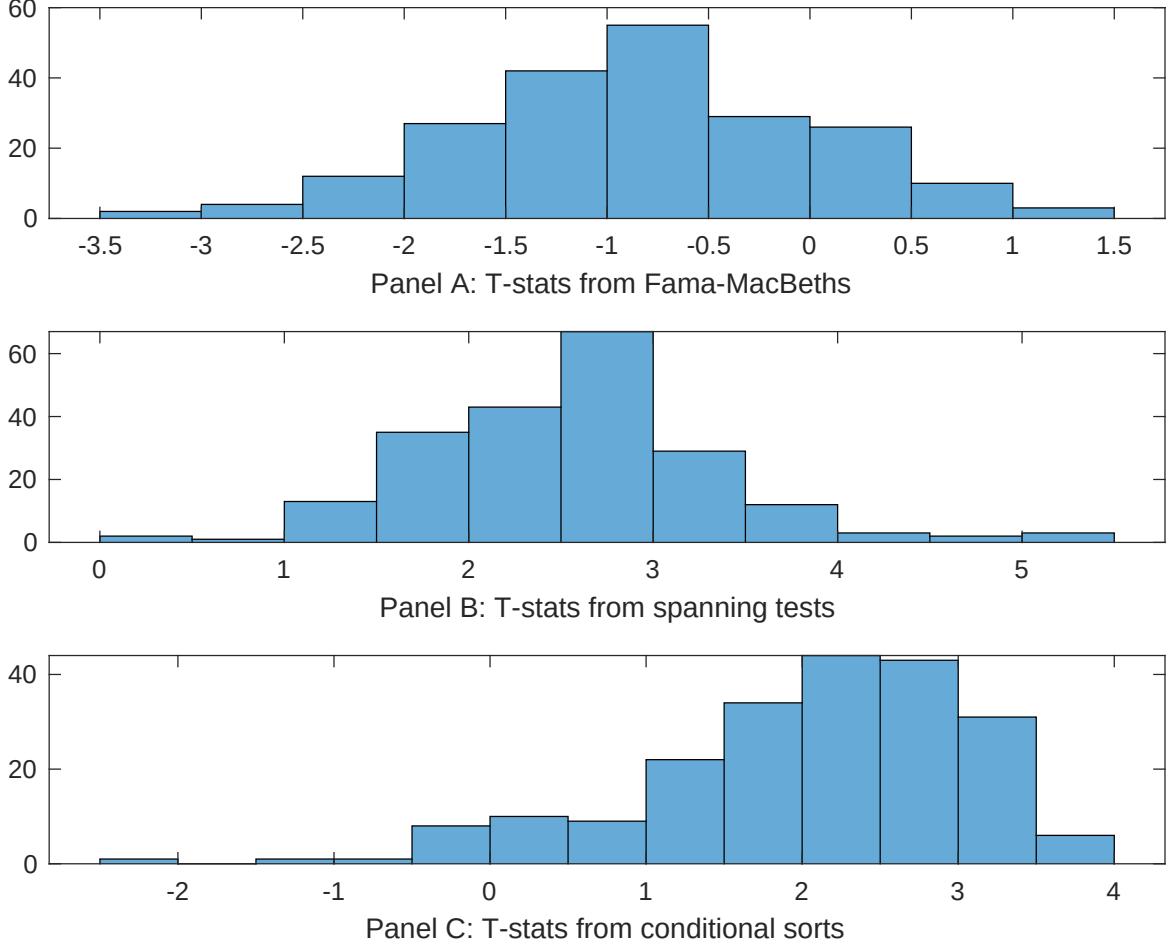


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RTS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RTS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RTS}RTS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RTS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RTS. Stocks are finally grouped into five RTS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RTS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RTS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RTS}RTS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Asset growth, Employment growth, Change in current operating assets, change in ppe and inv/assets, Growth in book equity, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.55]	0.12 [5.86]	0.13 [6.14]	0.14 [6.42]	0.17 [7.13]	0.12 [6.03]	0.13 [5.30]
RTS	0.34 [0.72]	-0.39 [-0.73]	0.21 [0.04]	0.36 [0.67]	-0.14 [-0.22]	-0.44 [-0.87]	0.57 [0.96]
Anomaly 1	0.97 [8.67]						0.54 [4.69]
Anomaly 2		0.83 [6.04]					0.30 [0.24]
Anomaly 3			0.18 [5.88]				0.57 [1.71]
Anomaly 4				0.15 [8.25]			0.66 [2.75]
Anomaly 5					0.42 [3.63]		-0.32 [-0.32]
Anomaly 6						0.14 [3.63]	-0.84 [-2.10]
# months	732	713	732	732	720	732	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RTS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RTS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Asset growth, Employment growth, Change in current operating assets, change in ppe and inv/assets, Growth in book equity, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.34 [4.31]	0.23 [2.84]	0.34 [4.27]	0.34 [4.36]	0.35 [4.36]	0.28 [3.48]	0.22 [2.80]
Anomaly 1	-24.35 [-5.13]						-12.40 [-2.20]
Anomaly 2		-34.37 [-7.98]					-23.70 [-5.13]
Anomaly 3			-19.11 [-4.89]				-2.04 [-0.45]
Anomaly 4				-25.46 [-6.94]			-16.19 [-4.14]
Anomaly 5					-4.88 [-1.11]		7.79 [1.63]
Anomaly 6						-21.96 [-5.85]	-8.39 [-1.84]
mkt	0.70 [0.37]	-0.14 [-0.07]	0.85 [0.45]	1.04 [0.55]	0.56 [0.29]	1.52 [0.80]	0.83 [0.45]
smb	10.89 [3.93]	7.49 [2.76]	4.95 [1.72]	9.96 [3.67]	9.10 [3.25]	8.29 [3.03]	8.68 [3.05]
hml	-8.17 [-2.22]	-3.24 [-0.87]	-1.83 [-0.46]	-6.22 [-1.71]	-8.44 [-2.24]	-0.55 [-0.14]	0.24 [0.06]
rmw	8.55 [2.31]	7.51 [2.05]	5.97 [1.60]	8.35 [2.28]	7.97 [2.11]	9.95 [2.69]	8.81 [2.41]
cma	-32.21 [-4.09]	-31.32 [-4.78]	-51.12 [-8.79]	-42.06 [-6.98]	-56.87 [-8.18]	-49.30 [-8.55]	-15.20 [-1.84]
umd	8.13 [4.30]	10.60 [5.66]	8.07 [4.25]	8.92 [4.80]	8.95 [4.66]	7.46 [3.94]	8.95 [4.80]
# months	732	713	732	732	732	732	713
$\bar{R}^2(\%)$	39	42	39	41	37	39	45

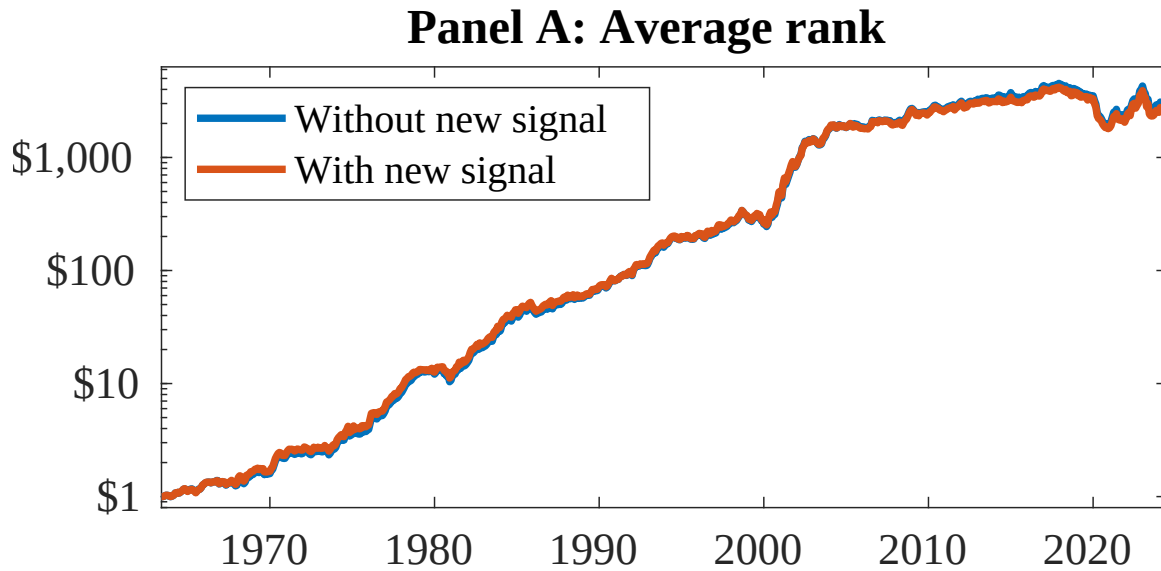


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as RTS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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